The same force can produce different accelerations

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Inverse relationship / ES: relación inversa / PT: relação inversa

Double M halves a at fixed net force.

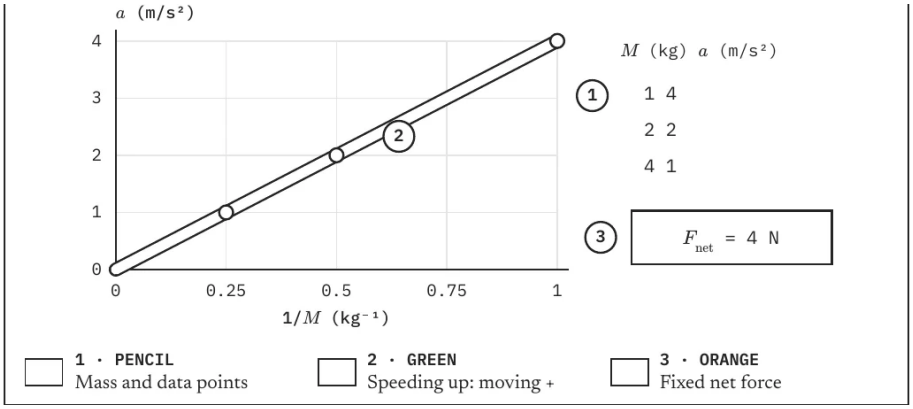
Also on your note page: System mass, Control, Slope.

B READ THE RELATIONSHIP

a = Fnet/M; the slope of a versus 1/M is Fnet

As you read, find: change · control · compare.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Under the same net force, a 1 kg system accelerates at 4 m/s² and a 2 kg system at 2 m/s² . Predict a for a 3 kg system.

Given M 1 = 1 kg ; a 1 = 4 m/s² M 2 = 2 kg ; a 2 = 2 m/s² same net force

Rule Check M × a . Then a = F net / M .

Work 1(4) = 2(2) = 4 N a 3 = 4 N ÷ 3 kg

Answer 1.33 m/s² Assumes net force stays 4 N .

E SAVE FOR THE FINAL MEETING

For the same force, a more massive XJ changes velocity less quickly. The comparison needs the same net force; a larger gravitational force can change that result.

Read the physics: Inverse relationship

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Compare a light cart with a loaded cart. Use the same net force and the same release method. Otherwise a change in acceleration could have more than one cause.

02 HOW IT WORKS

Read acceleration from the slope of a velocity–time graph. At fixed net force, doubling the moving mass halves acceleration. Multiplying mass by acceleration gives the same net force in both trials.

03 WHAT TO CHECK

Keep the hanging mass, track and release conditions consistent. Name what changed and what was controlled before stating the result.

04 TRY THE EXAMPLE FROM YOUR NOTES

Under the same net force, a 1 kg system accelerates at 4 m/s² and a 2 kg system at 2 m/s² . Predict a for a 3 kg system. 1.33 m/s² Assumes net force stays 4 N .

Model check: a = Fnet/M; the slope of a versus 1/M is Fnet.

05 CONNECT TO XJ

For the same force, a more massive XJ changes velocity less quickly. The comparison needs the same net force; a larger gravitational force can change that result.

SAY IT BEFORE YOU WRITE

Which quantity must match in the two trials? We changed ___. We kept ___ fixed. Acceleration ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

System mass

ES: masa del sistema / PT: massa do sistema

Meaning in my words	How can you tell?
How would you explain system mass to a friend? In my words, this means ___.	If a cart and hanging mass accelerate together, which masses belong in the total? I can tell because ___.
Example or sketch	**Non-example and why**
A loaded cart is 0.8 kg and its loaded hanger is 0.2 kg. What mass accelerates? My example shows ___.	Why leave the fixed track out of the moving mass? I would change ___ because ___.

Control

ES: variable controlada / PT: variável controlada

Meaning in my words	How can you tell?
How would you explain control to a friend? In my words, this means ___.	You test different cart masses. Why use the same hanging weight each time? I can tell because ___.
Example or sketch	**Non-example and why**
Name two things you would set the same before each release. My example shows ___.	A partner lists cart mass as a control while adding mass each trial. Where does it belong? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Inverse relationship

ES: relación inversa / PT: relação inversa

Meaning in my words	How can you tell?
How would you explain inverse relationship to a friend? In my words, this means ___.	Keep the net force fixed. If the moving mass doubles, what happens to acceleration? I can tell because ___.
Example or sketch	**Non-example and why**
At 4 N, compare the accelerations of carts of 1 kg and 2 kg. My example shows ___.	If both mass and net force double, must acceleration halve? I would change ___ because ___.

Slope

ES: pendiente / PT: inclinação

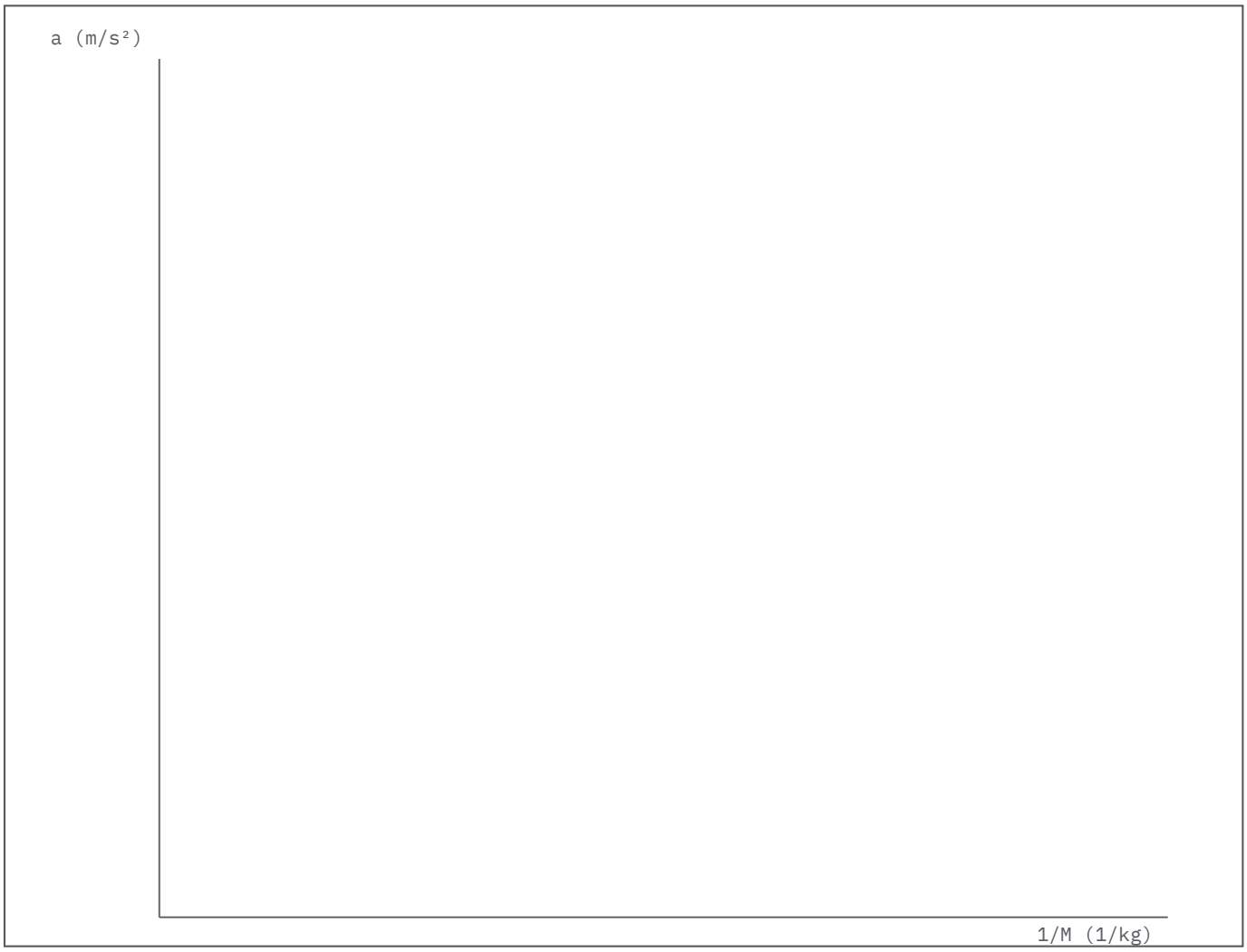
Meaning in my words	How can you tell?
How would you explain slope to a friend? In my words, this means ___.	At the same net force, which graph should be straight: a against M or a against 1/M? I can tell because ___.
Example or sketch	**Non-example and why**
Acceleration rises by 2 m/s² when 1/M rises by 0.5 per kg. What is the slope? My example shows ___.	Someone says this slope is the mass. How can you check? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a versus 1/M. Label the axes and units, draw the rising line, and name the quantity represented by its slope.

Labels you can use: change · control · compare.



Tell a partner

We changed ___. We kept ___ fixed. Acceleration ___.

Same push. Different mass.

11

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare velocity changes when masses differ.

CHOOSE ONE JOB TOGETHER

TEST ENGINEER ☐

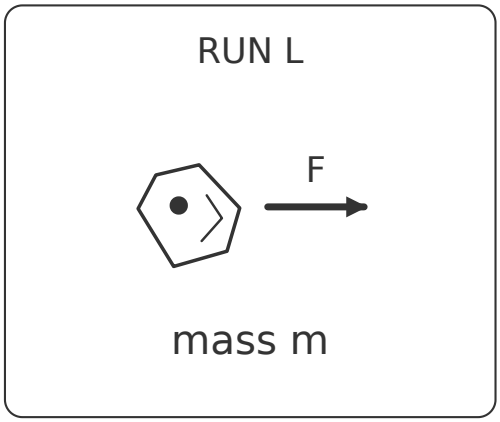
Compare the two test runs. Tell the crew which body changes velocity more and use the masses to explain why.

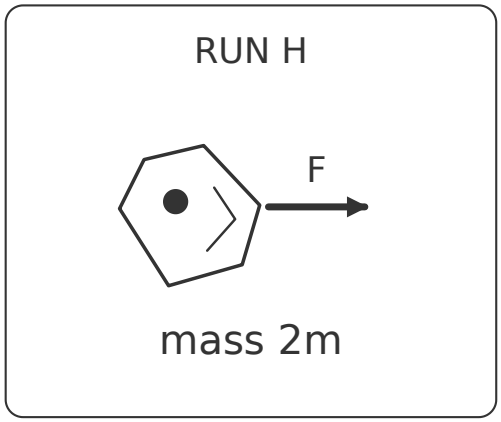
LAB CORRESPONDENT ☐

Explain why equal pushes can give different results. Have your partner check what stayed the same in both tests.

Word help: mass / net force / velocity change

Two asteroid models get the same net force for the same time. Model H has twice the mass of L.





01 OUR FIRST CALL

Which body changes velocity more? Explain your first call from the test card.

Run ___ changes more because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What changed in the test? Give the crew a new comparison.

Now we would say ___ because ___.

Switch speakers. Check both runs using mass times acceleration. Do your results match the forces on the test card?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks